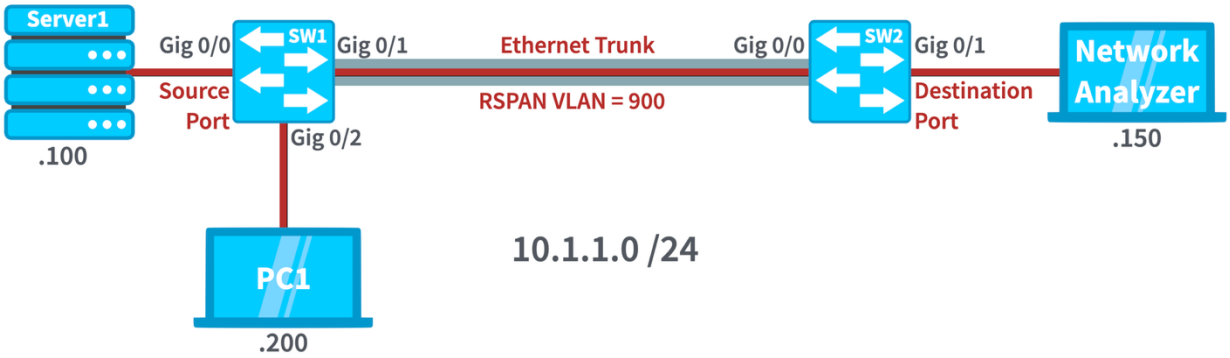
Remote SPAN Lab

Topology



Initial Configuration Commands

SERVER1 (Server Node in CML):

Paste the following under the node's CONFIG tab in CML:

```
# this is a shell script which will be sourced at boot
hostname SERVER1
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname SERVER1
sudo ifconfig eth0 10.1.1.100 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

PC1 (Desktop Node in CML):

Paste the following under the node's CONFIG tab in CML:

```
# this is a shell script which will be sourced at boot
hostname PC1
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname PC1
sudo ifconfig eth0 10.1.1.200 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

NETWORK-ANALYZER (Desktop Node in CML):

Paste the following under the node's CONFIG tab in CML:

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```
# this is a shell script which will be sourced at boot
hostname NETWORK-ANALYZER
# configurable user account
USERNAME=cisco
PASSWORD=cisco
sudo hostname NETWORK-ANALYZER
sudo ifconfig eth0 10.1.1.150 netmask 255.255.255.0 up
sudo route add default gw 10.1.1.1
```

SW1:

```
enable
conf t
host SW1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 4
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
int e0/0
switchport trunk encapsulation dot1q
switchport mode dynamic desirable
end
copy run star
```

SW2:

```
enable
conf t
host SW2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 4
password cisco
login
exec-timeout 0 0
```

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```
line con 0
logging synchronous
end
copy run star
```

Lab Task

1. Use the **`sudo tcpdump icmp`** command on the NETWORK-ANALYZER to monitor Ping traffic from PC1 to SERVER1 (10.1.1.100).
2. On PC1, issue the **`ping 10.1.1.100`** command to continuously Ping SERVER1.
3. Notice that no traffic is captured on the NETWORK-ANALYZER.
4. Configure an RSPAN VLAN of 900 on both switches.
5. Verify trunk configuration.
6. Configure SW1 to capture traffic in both directions on port e 0/1 and to send the mirrored traffic over the RSPAN VLAN.
7. Configure SW2 to take the captured traffic from the RSPAN VLAN and send it out port e 0/1 (i.e. to the Network Analyzer).
8. Notice that the NETWORK-ANALYZER is capturing ICMP traffic.

Solution

NETWORK-ANALYZER:

```
NETWORK-ANALYZER:~$ sudo tcpdump icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

PC1:

```
PC1:~$ ping 10.1.1.100
PING 10.1.1.100 (10.1.1.100): 56 data bytes
64 bytes from 10.1.1.100: seq=0 ttl=42 time=29.516 ms
```

NETWORK-ANALYZER:

[NOTICE NO TRAFFIC IS BEING CAPTURED]

SW1:

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 900
SW1(config-vlan)#remote-span
SW1(config-vlan)#end
SW1#show interfaces trunk

Port           Mode             Encapsulation  Status        Native vlan
Et0/0          desirable        802.1q         trunking      1
```

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```
Port           Vlans allowed on trunk
Et0/0          1-4094

Port           Vlans allowed and active in management domain
Et0/0          1,900

Port           Vlans in spanning tree forwarding state and not pruned
Et0/0          1,900
SW1#
```

SW2:

```
SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#vlan 900
SW2(config-vlan)#remote-span
SW2(config-vlan)#end
SW2#show interfaces trunk

Port           Mode             Encapsulation  Status        Native vlan
Et0/0          auto             n-802.1q       trunking      1

Port           Vlans allowed on trunk
Et0/0          1-4094

Port           Vlans allowed and active in management domain
Et0/0          1,900

Port           Vlans in spanning tree forwarding state and not pruned
Et0/0          1,900
SW2#
```

SW1:

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#monitor session 1 source interface e0/1
SW1(config)#monitor session 1 destination remote vlan 900
SW1(config)#end
SW1#show monitor session 1
Session 1
---------
Type                    : Remote Source Session
Source Ports            :
    Both                : Et0/1
Dest RSPAN VLAN         : 900


SW1#
```

SW2:

```
SW2#conf t
```

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```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#monitor session 1 source remote vlan 900
SW2(config)#monitor session 1 destination interface e0/1
SW2(config)#end
SW2#show monitor session 1
Session 1
---------
Type                      : Remote Destination Session
Source RSPAN VLAN         : 900
Destination Ports         : Et0/1
    Encapsulation         : Native


SW2#
```

NETWORL-ANALYZER:

```
20:06:08.940339 IP 10.1.1.100 > 10.1.1.200: ICMP echo reply, id 1, seq 445,
length 64
20:06:09.940376 IP 10.1.1.200 > 10.1.1.100: ICMP echo request, id 1, seq 446,
length 64
20:06:09.941279 IP 10.1.1.100 > 10.1.1.200: ICMP echo reply, id 1, seq 446,
length 64
```

[NOTICE ICMP TRAFFIC IS BEING CAPTURED]

Also, just like we did with Local SPAN, on switch SW1 we could have made our source a VLAN rather than an interface.

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